

Report File

Project Title: Student Performance Analysis: Data Cleaning and Visualization Using Python

Name: Ansh Gautam

Abstract

This project presents a comprehensive data analysis of student performance using computational techniques in Python. Leveraging the Student Performance Dataset from the UCI Machine Learning Repository, the study examines 649 student records across 33 demographic, social, and academic attributes to identify key factors influencing final academic outcomes. Through rigorous data preprocessing using the Pandas library, the dataset was verified to possess excellent structural integrity, containing zero missing values and zero duplicate records. Exploratory Data Analysis (EDA) was subsequently conducted using Matplotlib and Seaborn to compress complex multidimensional relationships into interpretable visual models. The analysis generated targeted visualizations, including histograms, box plots, scatter plots, and a correlation heatmap. The findings indicate that early academic benchmarks—specifically first and second-period grades—are the strongest predictors of final success. Furthermore, the study identifies a law of diminishing marginal returns concerning weekly study time and highlights the tangible academic penalty associated with a lack of home internet access and high absenteeism. Ultimately, this report underscores the value of data-driven decision-making in educational institutions to facilitate early interventions and equitable resource allocation.

1. Introduction

The intersection of education and computational statistics represents a transformative frontier in modern pedagogy. Historically, academic institutions have relied on reactive measures to gauge student success, often waiting until final examinations to identify which individuals were struggling. However, the advent of data analytics allows for a fundamental shift toward predictive and preventative strategies.

Data cleaning and visualization are foundational pillars of this analytical process. Raw educational data is frequently unstructured or fraught with inconsistencies. By applying rigorous data cleaning methodologies, analysts ensure that the underlying data is mathematically valid, adhering to the principle of "Garbage In, Garbage Out." Subsequent data visualization acts as the bridge between complex statistical matrices and actionable human insight, translating millions of data points into pre-attentive visual geometries that stakeholders can instantly comprehend.

The primary purpose of this project is to apply Python-based data cleaning and visualization techniques to a real-world educational dataset. By systematically decoding the demographic, social, and behavioural factors that correlate with academic outcomes, this project aims to provide empirical evidence that can guide institutional decision-making and pedagogical support strategies.

2. Dataset Description

The empirical foundation of this analysis is the "Student Performance Dataset," sourced from the UCI Machine Learning Repository. This dataset tracks the academic achievement of secondary education students across two distinct Portuguese schools. While the repository offers data for both Mathematics and Portuguese courses, this specific project utilizes the student-por.csv file, which evaluates performance exclusively within the Portuguese language course.

The dataset comprises 649 distinct records (rows) and 33 attributes (columns). These variables capture a holistic view of the student's ecosystem. Key columns analysed in this project include:

- **sex:** A categorical binary variable indicating the biological sex of the student (F for Female, M for Male).
- **study time:** An ordinal numerical variable measuring weekly extracurricular study time (1: <2 hours, 2: 2-5 hours, 3: 5-10 hours, 4: >10 hours).
- **absences:** A discrete numerical variable representing the cumulative number of school days the student was absent.
- **internet:** A categorical binary variable indicating the availability of internet access within the student's home environment (yes or no).
- **G1:** A continuous numerical variable representing the first-period assessment grade (scaled 0 to 20).
- **G2:** A continuous numerical variable representing the second-period assessment grade (scaled 0 to 20).
- **G3:** A continuous numerical variable representing the final summative grade (scaled 0 to 20). This serves as the primary dependent variable for our analysis.

3. Methodology

The methodology for this project relies on a programmatic data science pipeline executed in Python. The workflow is divided into two primary phases: Data Preprocessing and Exploratory Data Analysis (EDA).

Data Loading using Pandas: The project commenced by ingesting the raw CSV file into the Python environment. Utilizing the Pandas library, the data was parsed into a two-dimensional Data Frame architecture, enabling programmatic manipulation and statistical assessment.

Missing Value Checking: To ensure mathematical validity, the dataset was scanned for undefined, blank, or NaN (Not a Number) entries using the `df.isnull().sum()` function. The algorithm confirmed that the dataset was entirely complete, returning 0 missing values across all 33 columns.

Duplicate Checking: The matrix was subsequently evaluated for clerical double-entry errors using the `df.duplicated().sum()` function. The scan confirmed 0 duplicate records within the 649 rows.

Data Preprocessing: Although the dataset was inherently pristine, robust data engineering principles were applied. The `df.drop_duplicates(inplace=True)` command was programmatically executed as a structural safeguard. The verified, sanitized Data Frame was then successfully exported to a new file, `cleaned_student_data.csv`, to serve as the immutable source for all visualization tasks.

Visualization Process: To conduct Exploratory Data Analysis, the Matplotlib and Seaborn libraries were deployed. Matplotlib was utilized to structure the underlying geometry and Cartesian axes of the charts, while Seaborn was leveraged to map statistical aggregations (such as medians and quartiles) and apply distinct colour palettes, ultimately generating six specialized visual models.

4. Results and Analysis

4.1 Distribution of Final Grades (Histogram)

What the graph shows: A histogram grouping the continuous G3 final grades into discrete bins along the X-axis, with the Y-axis representing the frequency of students within each grade bracket. **Insights obtained:** The distribution of grades approximates a normal bell curve, with a pronounced central mode tightly clustered in the passing range of 10 to 13. This indicates a generally healthy academic cohort. A structural anomaly—an isolated spike at the exact zero coordinate—highlights a distinct subgroup of students. Rather than representing a 0% academic assessment, this cluster identifies administrative dropouts, complete trancies during finals, or disqualifications.

4.2 Gender vs Final Grade (Box Plot)

What the graph shows: A box-and-whisker plot comparing the categorical variable of gender (sex) against the continuous final grade (G3). **Insights obtained:** The visual geometry demonstrates that female students achieve a median final grade of 12, slightly outperforming male students who sit at a median of 11. Furthermore, the upper quartile boundary is visibly higher for the female cohort, indicating that the top 25% of female students score higher than their male counterparts in this language course.

4.3 Study Time vs Final Grade (Box Plot)

What the graph shows: A box plot mapping the ordinal categories of weekly study time against the final G3 grade. **Insights obtained:** As expected, increased study time generally yields higher grades; however, the graph visualizes the economic law of diminishing marginal returns. While moving from Category 1 (<2 hours) to Category 3 (5-10 hours) results in a steady climb in median grades (from 11 to 13), transitioning to Category 4 (>10 hours) shows a flatline. The median grade ceases to rise significantly, suggesting that excessive studying without improved methodology leads to fatigue rather than proportional academic gains.

4.4 Internet Access vs Final Grade (Box Plot)

What the graph shows: A box plot comparing the binary home internet access variable against the final G3 grade. **Insights obtained:** The graph serves as a proxy indicator for digital equity. Students with home internet access display a firm median grade of 12 and a higher overall grading ceiling. Conversely, students lacking internet access exhibit a suppressed median grade of 11 and a lower maximum score, visualizing the tangible academic penalty imposed by the digital divide.

4.5 Absences vs Final Grade (Scatter Plot)

What the graph shows: A scatter plot mapping the exact Cartesian coordinates of a student's total absences (X-axis) against their G3 final grade (Y-axis). **Insights obtained:** The plot reveals a negative correlation trajectory. A dense cluster of high-performing students exists between 0 and 5 absences. As the eye tracks right toward higher absenteeism (15+ days), the data points disperse and uniformly sink toward the bottom half of the grading spectrum. While a few outliers survive high truancy, the vast majority suffer severe academic degradation.

4.6 Correlation Heatmap

What the graph shows: A multivariate colour matrix visualizing the Pearson Correlation Coefficients for all numerical variables in the dataset. Deep red indicates strong positive

correlation, while deep blue indicates strong negative correlation. **Insights obtained:** The heatmap provides the most potent mathematical evidence in the study. The interim grades, G1 and G2, exhibit blindingly strong positive correlations with the final G3 grade (0.83 and 0.92, respectively). Additionally, the variable failures show the strongest negative correlation (-0.39) with the final grade, indicating that a history of academic struggle heavily predicts future failure.

5. Key Findings

Based on the computational analysis and visual evidence, the major observations are:

- **Study Time Efficacy:** Students who spend more time studying generally achieve higher final grades, though effort beyond 10 hours weekly shows diminishing returns.
- **The Digital Divide:** Students with home internet access tend to perform slightly better academically, highlighting the importance of socio-technological resources.
- **The Truancy Penalty:** Students with higher absenteeism generally achieve lower grades, establishing a clear link between classroom presence and academic output.
- **Predictive Power of Early Grades:** Previous academic performance indicators (G1 and G2) are overwhelmingly related to final grades (G3), proving that a student's trajectory is established early in the academic year.
- **Data Integrity:** The dataset provided was exceptionally clean, requiring no imputation for missing values or deletion of duplicate records.

6. Conclusion

This project successfully executed a data cleaning and visualization pipeline to decode the underlying mechanics of student performance. By transforming raw CSV data into comprehensive visual models, the analysis isolated the specific academic, behavioural, and demographic variables that most heavily influence a student's final grade.

The outcomes of this project underscore the critical importance of data-driven decision-making in the educational sector. Rather than relying on intuition or broad disciplinary measures, educators can use these empirical findings to enact targeted changes. Recognizing that early assessment grades perfectly predict final failure allows institutions to shift tutoring resources to the beginning of the semester. Acknowledging the academic tax of the digital divide encourages schools to invest in hardware or internet access programs. Ultimately, analysing data shifts education from a system of reactive grading to one of proactive student support.

7. Future Scope

While Exploratory Data Analysis provides deep diagnostic insights, the next evolutionary step for this project is predictive modelling.

Machine Learning Prediction of Student Grades: Future iterations of this project will utilize the scikit-learn library to deploy supervised Machine Learning algorithms—such as Linear Regression, Random Forest Regressors, or Support Vector Machines (SVM). By training a model on the historical dataset, the system will be able to ingest a new student's profile (their age, study time, absences, and G1 grade) and mathematically predict their precise G3 final grade before the semester even concludes.

Predictive Analytics and Educational Interventions: By integrating Machine Learning predictions into a school's administrative dashboard, educators can be automatically flagged when a student's predictive trajectory dips below a passing threshold. This facilitates automated, real-time educational interventions, allowing counsellors to deploy personalized support plans precisely when the student needs them most.

References

1. **Pandas Documentation:** Data manipulation and analysis library. Available at: <https://pandas.pydata.org/docs/>
2. **Matplotlib Documentation:** Comprehensive library for creating static, animated, and interactive visualizations in Python. Available at: <https://matplotlib.org/stable/contents.html>
3. **Seaborn Documentation:** Statistical data visualization library. Available at: <https://seaborn.pydata.org/>
4. **UCI Machine Learning Repository:** Student Performance Dataset. Paulo Cortez, University of Minho, Guimarães, Portugal. Available at: <https://archive.ics.uci.edu/dataset/320/student+performance>